

Sunmook Choi, 최선목

<https://smfelixchoi.github.io>

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RESEARCH INTERESTS *I study bandits, reinforcement learning, optimal control using tools and concepts from statistical learning theory, optimization, and control theory.*

EDUCATION **Cornell University**

- Ph.D. student, Center for Applied Mathematics Aug. 2024 – Present
- Ph.D. Minors: Mathematics, Computer Science
- Thesis Committee: Sarah Dean (Advisor), David Bindel (Math), Robert Kleinberg (CS)

Korea University

- M.S., Mathematics Mar. 2022 – Aug. 2024
Thesis: Three-dimensional dicube tilings of augmented Aztec bipyramids
Advised by Prof. Seungsang Oh
- B.S., Mathematics Mar. 2016 – Feb. 2022
- Korean Mandatory Military Service Jun. 2017 – Feb. 2019

- HONORS & AWARDS
- KUAA 2024 Scholarship (Korea University Alumni Association in NY) Dec 2024
 - [Editors' Choice Award 2023, Discrete Mathematics](#) Mar 2024
 - IJCAI 2023 Competition and Challenge Award Aug 2023
 - ICASSP 2022 Grand Challenge Award Mar 2022
 - Sigma Scholarship (Korea Univ. Dept. of Mathematics Alumni Association) Fall 2021
 - Dean's List, College of Science, Korea University Fall 2020
 - Semester High Honors, Korea University Spring (2016,2019,2020,2021), Fall (2019,2020,2021)
 - SAMSUNG Welfare Foundation Scholarship (Dream Class) Spring 2019

SELECTED PUBLICATIONS A full list of papers can be found in [Google Scholar](#).

Preprint

- [1] *Dual Control of Linear Systems from Bilinear Observations with Belief Space Model Predictive Control*.
arXiv:2604.24663
Daniel Cao, Beixi Du, Andrew Lowitt, **Sunmook Choi**, Sarah Dean, Yahya Sattar

Conference Papers

- [1] *Two-Layer Linear Auto-Regressive Models Estimate Latent States*
International Conference on Machine Learning (ICML), 2026
Yahya Sattar, **Sunmook Choi**, Leo Maynard-Zhang, Yassir Jedra, Maryam Fazel, Sarah Dean
- [2] *Explore-then-Commit for Nonstationary Linear Bandits with Latent Dynamics*
International Conference on Artificial Intelligence and Statistics (AISTATS), 2026
Sunmook Choi, Yahya Sattar, Yassir Jedra, Maryam Fazel, Sarah Dean
- [3] *Sub-optimality of the Separation Principle for Quadratic Control from Bilinear Observations*
IEEE Conference on Decision and Control (CDC), 2025
Yahya Sattar, **Sunmook Choi**, Yassir Jedra, Maryam Fazel, Sarah Dean
- [4] *TB-ResNet: Bridging the gap from TDNN to ResNet in Speaker Verification with Temporal-Bottleneck Enhancement*
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2024
Sunmook Choi, Sanghyeok Chung, Seungeun Lee, Soyul Han, Taein Kang, Jaejin Seo, Il-Youp Kwak, Seungsang Oh
- [5] *Overlapped Frequency-Distributed Network: Frequency-Aware Voice Spoofing Countermeasure*
Interspeech, 2022
Sunmook Choi, Il-Youp Kwak, Seungsang Oh
- [6] *Light-weight Frequency Information Aware Neural Network Architecture for Voice Spoofing Detection*
International Conference on Pattern Recognition (ICPR), 2022
Sunmook Choi, Seungsang Oh, Jonghoon Yang, Yerin Lee, Il-Youp Kwak

Workshop Papers

- [1] *CAU KU deep fake detection system for ADD 2023 challenge*
Workshop on Deepfake Audio Detection and Analysis at IJCAI, 2023
Soyul Han*, Taein Kang*, **Sunmook Choi***, Jaejin Seo, Sanghyeok Chung, Seungeun Lee, Seungsang Oh, Il-Youp Kwak
(*: equal contribution)
- [2] *Low-quality Fake Audio Detection through Frequency Feature Masking*
International Workshop on Deepfake Detection for Audio Multimedia at ACM Multimedia, 2022
Il-Youp Kwak, **Sunmook Choi**, Jonghoon Yang, Yerin Lee, Soyul Han, Seungsang Oh

Journal Articles

- [1] *Augmented Aztec bipyramid and dicube tilings*
Sunmook Choi, Sangyop Lee, Seungsang Oh
Discrete Mathematics, 2024, **Editor's Choice**

Thesis

- [1] *Three-dimensional dicube tilings of augmented Aztec bipyramids*
Master's thesis, Korea University [DOI link](#)

TEACHING

Teaching Assistant, Dept. of Mathematics, Cornell University

- Linear Algebra (MATH 4310) Spring 2026
- Linear Algebra for Engineers (MATH 2940) Fall 2025
- Linear Algebra for Engineers (MATH 2940) Spring 2025
- Linear Algebra for Engineers (MATH 2940) Fall 2024

Teaching Assistant, [Dept. of Mathematics, Korea University](#)

- Linear Algebra II With LAB (MATH 222) Fall 2023
- Linear Algebra I With LAB (MATH 221) Spring 2023
- Linear Algebra II With LAB (MATH 222) Fall 2022
- Linear Algebra I With LAB (MATH 221) Spring 2022

SERVICE & LEADERSHIP

Reviewer: NeurIPS, TMLR, ICML

Research Mentor, [The Bowers Undergraduate Research Experience \(BURE\) Program](#) Summer 2026

Reading Mentor, [Directed Reading Program, Math Department, Cornell](#) Spring 2026

- Guided an undergraduate student through mathematical reading and weekly discussions

Project Mentor, M.Eng. Program, Cornell Fall 2025 – Spring 2026

- Guided a student project undertaken for academic credit.

Program Mentor, CURIE Academy, Cornell Engineering High School Outreach Program Summer 2025

- Control & Navigation of Weather Balloons, led by Prof. Sarah Dean

Seminar Leader, Deep Learning Seminar, Season 1 & 2, [DLmath](#) Jun. 2023 – Aug. 2023, Sep. 2022 – Feb. 2023

- Theoretical Review and Python Implementation of SGD, MLP, CNN, RNN, AE, VAE, GAN.
- [Season 1 materials](#), [Season 2 materials](#)

Seminar Leader, Deep Reinforcement Learning Seminar, Mar. 2023 – Sep. 2023

- Theoretical Review and Python Implementation of MDP, DP, RL, DQN, PG, AC, PPO.
- [Materials](#)

Tutor, SAMSUNG Dream Class Spring 2019

- Teaching Mathematics to middle school students as an after-school program.

COMPETENCES

Languages Korean, English

Techniques $\LaTeX$, Python, TensorFlow, PyTorch, R, Julia